

Mehmet Aygun

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RESEARCH INTERESTS

I am interested in computer vision and machine learning. In particular, I am interested in developing image understanding models that can learn fundamental priors and representations through self-supervision alone. Examples of these include developing models for single-view 3D object reconstruction and investigating the role of shape in semantic understanding tasks.

EDUCATION

- **The University of Edinburgh (UoE), School of Informatics** Edinburgh, UK
Doctor of Philosophy in Computer Science, *Sep. 2021 – Current*
- **Technical University of Munich (TUM), Department of Informatics** Munich, Germany
Master of Science in Computer Science *Oct. 2018 – Dec. 2020*
- **Istanbul Technical University (ITU)** Istanbul, Turkey
Bachelor of Science in Computer Science *Sep. 2013 – June. 2017*

RESEARCH EXPERIENCE

- **Institute for Adaptive and Neural Computation at UoE** Edinburgh, UK
Postgraduate Research Student *Sep 2021 - Current*
 - **Shape & Semantics:** Developing models for single-view 3D reconstruction for articulated objects and exploring the role of shape in semantic tasks like correspondence and fine-grained classification. Advisor: Dr. Oisín Mac Aodha
- **TUM Computer Vision Lab** Munich, DE
Research Student *Oct 2018 - Jun 2021*
 - **Holistic Scene Analysis:** Worked on semantic/instance segmentation and multi object tracking with Lidar data. Advisors: Prof. Laura Leal-Taixe, Dr. Aljosa Osep.
 - **3D Shape Analysis:** Worked on shape correspondence problem with self-supervised deep learning methods using functional maps and heat kernels. Advisors: Prof. Daniel Cremers, Dr. Zorah Lahner.
- **ITU Computer Vision Lab** Istanbul, Turkey
Research Student *Dec 2013 - June 2017*
 - **Medical Image Segmentation:** Worked on multi-modal 3D Brain Tumor segmentation using convolutional neural networks under supervision of Prof. Gozde Unal.
 - **Transfer Learning:** Worked on transfer learning for convolutional neural networks. Collaborated with Dr. Yusuf Aytar from MIT CSAIL and proposed a statistical regularization method for transferring knowledge between different CNN architectures.

PUBLICATIONS

- **M. Aygun**, O. Mac Aodha, “Demystifying Unsupervised Semantic Correspondence Estimation”, In European Conference on Computer Vision (ECCV), 2022
- **M. Aygun**, A. Osep, M. Weber, M. Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, “4D Panoptic Lidar Segmentation”, In Conference on Computer Vision and Pattern Recognition (CVPR), 2021
- **M. Aygun**, Z. Lahner, D. Cremers, “Unsupervised Dense Shape Correspondence using Heat Kernels”, In International Conference on 3D Vision (3DV), 2020
- **M. Aygun**, Y. Aytar, H. K. Ekenel, “Exploiting Convolution Filter Patterns for Transfer Learning”, In ICCV TASK-CV workshop, 2017. (**Honourable Mention Award**)
- R. C. Malli*, **M. Aygun***, H. K. Ekenel, “Apparent Age Estimation Using Ensemble of Deep Learning Models”, In CVPR Workshop, 2016 (*:equal contribution.)

HONORS & AWARDS

- **ICCV 17 Workshop, Honourable Mention Award:** Received Honourable Mention Award at ICCV 17 TASK-CV workshop for our paper “Exploiting Convolution Filter Patterns for Transfer Learning”.
- **DAAD scholarship:** Two years of master scholarship for studying in Germany.
- **Bachelor Thesis Award:** Best Bachelor Thesis Award from Computer Engineering Department for my thesis titled “Multi Modal 3D Convolutional Neural Networks for Brain Tumor Segmentation”
- **Undergraduate Research Scholarship :** Research scholarship from science and technology research council of Turkey during my last two year of bachelor education.

PROFESSIONAL EXPERIENCE

- **Coriolis Pharma** Munich, Germany
Computer Vision Engineer *Sep 2019 - March 2020*
 - **Biological image analysis:** Developed methods for particle analysis using deep learning based classification, instance segmentation and unsupervised learning methods. Used frameworks: Tensorflow, Keras, OpenCV.
- **Motius** Munich, Germany
Machine Learning Engineer *Jan 2019 - April 2019*
 - **Handwritten Text Recognition:** Developed handwritten text recognition system using deep learning (CNN+LSTM+CTC). Reduced word error rate from 18% to 12%. Used frameworks: Tensorflow, OpenCV.
- **Bunsar** Istanbul, Turkey
Computer Vision Engineer *Feb 2017 - Sep 2018*
 - **Visual Search Engine from Images and Videos:** Research and development activities for image retrieval system using deep learning methods. Topic studied: Object Detection, Image Segmentation, Image Classification, Large Scale Image Retrieval Used frameworks: Caffe, OpenCV, Sklearn

SKILLS

- Python, C/C++, PyTorch{3D}, OpenCV, Numpy, Matlab, Linux, Git

RELATED COURSEWORK

- **Technical University of Munich:** Natural Language Processing, Introduction to Deep Learning, Variational Methods for Computer Vision, Multiple View Geometry, Tracking and Detection
- **Istanbul Technical University:** Introduction to Computer Vision, Learning from Data, Probability, Linear Algebra, Calculus I & II

ACTIVITIES

- Reviewer for CVPR23, CVPR22, ECCV22, 3DV22, FGVC9
- Tutoring for Introduction to ML Course at UoE 2021, 2022
- Volunteering work at code.org for high school students between 2015-2017

REFERENCES

- **Dr. Oisin Mac Aodha:** oisin.macaodha@ed.ac.uk
- **Prof. Laura Leal-Taixe:** leal.taixe@tum.de
- **Prof. Gozde Unal:** gozde.unal@itu.edu.tr